Molecular Modeling in Process Engineering

2025/2026

Project 8

MEP estimation based on a multireference method

It is required to estimate the MEP for the reaction:

$CH_4 \rightarrow CH_3 + H$

Calculations must be performed with CASPT2 as the level of theory and cc-pVDZ as the basis set.
One active space must be chosen.
Report on excel the energy trend at different C-H distances and compare the values obtained with CASPT2 and those obtained at the HF level.

The use of Molpro as software is recommended. Where in the output file is the information about the basis set used? and the energy of the different types of interaction?